

一、前言

1. 磁学简述.

2. 静磁场与静电场的比较.

二、小线圈问题

小线圈：线圈尺度 $\ll$ 到场点的距离.

$$r' \ll r$$

对于均匀磁场，任何线圈都是小线圈.

★ 小线圈的磁场、磁矢势

$$\vec{B} = -\frac{\mu_0 \vec{m}}{4\pi r^3} + \frac{3\mu_0 (\vec{m} \cdot \vec{r})}{4\pi r^5} \vec{r}$$

$$\vec{A} = \frac{\mu_0 \vec{m} \times \vec{r}}{4\pi r^3}$$

★ 受磁场作用力、力矩

$$\vec{F} = (\vec{m} \cdot \nabla) \vec{B}$$

$$\vec{\tau} = \vec{m} \times \vec{B} + \vec{r} \times (\vec{m} \cdot \nabla) \vec{B}$$

小线圈唯一的量为磁矩 $\vec{m}$ ，定义为

$$\vec{m} = I \vec{S} = I \oint_L \frac{1}{2} \vec{r}' \times d\vec{r}'$$

例1. 求小线圈的磁场

$$\vec{R} = \vec{r} - \vec{r}', \quad R^2 = r^2 + r'^2 - 2\vec{r} \cdot \vec{r}'$$

$$r' \ll r$$

$$\Rightarrow R \approx r - \frac{\vec{r} \cdot \vec{r}'}{r}$$

$$\frac{1}{R^3} \approx \frac{1}{r^3} \left(1 + 3 \frac{\vec{r} \cdot \vec{r}'}{r^2} \right)$$

$$\frac{\vec{R}}{R^3} \approx \frac{1}{r^3} \left(1 + 3 \frac{\vec{r} \cdot \vec{r}'}{r^2} \right) (\vec{r} - \vec{r}') \approx \frac{1}{r^3} \vec{r} - \frac{\vec{r}'}{r^3} + \frac{3(\vec{r} \cdot \vec{r}') \vec{r}}{r^5}$$

$$\begin{aligned} \Rightarrow \vec{B} &= \frac{\mu_0}{4\pi} \oint_L \frac{I d\vec{r}' \times \vec{R}}{R^3} = \frac{\mu_0 I}{4\pi} \oint_L d\vec{r}' \times \frac{\vec{R}}{R^3} \\ &\approx \frac{\mu_0 I}{4\pi} \left[\oint_L d\vec{r}' \times \frac{\vec{r}}{r^3} - \oint_L d\vec{r}' \times \frac{\vec{r}'}{r^3} + \oint_L d\vec{r}' \times \frac{3(\vec{r} \cdot \vec{r}') \vec{r}}{r^5} \right] \\ &= \frac{\mu_0 I}{4\pi} \left[-\frac{\vec{r}}{r^3} \times \oint_L d\vec{r}' + \frac{1}{r^3} \oint_L \vec{r}' \times d\vec{r}' - \frac{3\vec{r}}{r^5} \times \oint_L (\vec{r} \cdot \vec{r}') d\vec{r}' \right] \end{aligned}$$

$$d((\vec{r} \cdot \vec{r}') \vec{r}') = (\vec{r} \cdot \vec{r}') d\vec{r}' + (\vec{r} \cdot d\vec{r}') \vec{r}'$$

$$(\vec{r}' \times d\vec{r}') \times \vec{r} = (\vec{r} \cdot \vec{r}') d\vec{r}' - (\vec{r} \cdot d\vec{r}') \vec{r}'$$

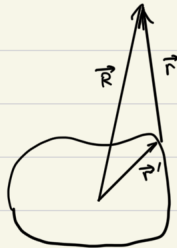
$$\Rightarrow (\vec{r} \cdot \vec{r}') d\vec{r}' = \frac{1}{2} \left[d((\vec{r} \cdot \vec{r}') \vec{r}') + (\vec{r}' \times d\vec{r}') \times \vec{r} \right]$$

$$\Rightarrow \vec{B} = \frac{\mu_0}{4\pi} \left[0 + \frac{2\vec{m}}{r^3} - I \frac{3\vec{r}}{r^5} \times \left(\oint_L \frac{1}{2} (\vec{r}' \times d\vec{r}') \times \vec{r} \right) \right]$$

$$= \frac{\mu_0}{4\pi} \left[\frac{2\vec{m}}{r^3} - \frac{3\vec{r} \times (\vec{m} \times \vec{r})}{r^5} \right]$$

$$= \frac{\mu_0}{4\pi} \left[\frac{2\vec{m}}{r^3} - \frac{3\vec{r} \cdot \vec{m} - (\vec{m} \cdot \vec{r}) \vec{r}}{r^5} \right]$$

$$= -\frac{\mu_0 \vec{m}}{4\pi r^3} + \frac{3\mu_0 (\vec{m} \cdot \vec{r})}{4\pi r^5} \vec{r}$$



例2. 矢势的多极展开.

$$\frac{1}{|\vec{r} - \vec{r}'|} \approx \frac{1}{r} - \vec{r}' \cdot \nabla \frac{1}{r} + \frac{1}{2!} \sum_{i,j} r'_i r'_j \frac{\partial^2}{\partial r_i \partial r_j} \left(\frac{1}{r} \right) + \dots$$

$$\vec{A} = \frac{\mu_0}{4\pi} \iiint_V \frac{\vec{j}(\vec{r}') dV'}{|\vec{r} - \vec{r}'|}$$

$$\approx \frac{\mu_0}{4\pi} \iiint_V \vec{j}(\vec{r}') \left(\frac{1}{r} - \vec{r}' \cdot \nabla \frac{1}{r} + \dots \right) dV'$$

$$\vec{A}^{(0)} = \frac{\mu_0}{4\pi r} \iiint_V \vec{j}(\vec{r}') dV' = 0$$

$$\vec{A}^{(1)} = \frac{\mu_0}{4\pi r^3} \iiint_V \vec{j}(\vec{r}') \vec{r} \cdot \vec{r}' dV'$$

$$\begin{aligned} \vec{r} \times (\vec{r}' \times \vec{j}(\vec{r}')) &= (\vec{r} \cdot \vec{j}(\vec{r}')) \vec{r}' - (\vec{r}' \cdot \vec{r}) \vec{j}(\vec{r}') \\ &= -2(\vec{r} \cdot \vec{r}') \vec{j}(\vec{r}') \end{aligned}$$

$$\begin{aligned} \Rightarrow \vec{A}^{(1)} &= -\frac{\mu_0}{4\pi r^3} \frac{1}{2} \iiint_V \vec{r} \times (\vec{r}' \times \vec{j}(\vec{r}')) dV' \\ &= \frac{\mu_0}{4\pi r^3} \left(\iiint_V \frac{1}{2} \vec{r}' \times \vec{j}(\vec{r}') dV' \right) \times \vec{r} \\ &= \frac{\mu_0 \vec{m} \times \vec{r}}{4\pi r^3} \end{aligned}$$

例3. 均匀磁场 $\vec{B}$ 中小线圈受到磁场作用力、力矩.

$$\vec{F} = \oint_L I d\vec{r} \times \vec{B} = I \oint_L d\vec{r} \times \vec{B} = 0$$

$$\vec{\tau} = \oint_L \vec{r} \times (I d\vec{r} \times \vec{B}) = I \oint_L \vec{r} \times (d\vec{r} \times \vec{B})$$

$$(\vec{r} \times d\vec{r}) \times \vec{B} = (\vec{r} \cdot \vec{B}) d\vec{r} - (d\vec{r} \cdot \vec{B}) \vec{r}$$

$$d((\vec{r} \cdot \vec{B}) \vec{r}) = (d\vec{r} \cdot \vec{B}) \vec{r} + (\vec{r} \cdot \vec{B}) d\vec{r}$$

$$\Rightarrow (\vec{r} \times d\vec{r}) \times \vec{B} + d((\vec{r} \cdot \vec{B}) \vec{r}) = 2(\vec{r} \cdot \vec{B}) d\vec{r}$$

$$\begin{aligned} \Rightarrow \vec{r} \times (d\vec{r} \times \vec{B}) &= (\vec{r} \cdot \vec{B}) d\vec{r} - (\vec{r} \cdot d\vec{r}) \vec{B} \\ &= \frac{1}{2} (\vec{r} \times d\vec{r}) \times \vec{B} + \frac{1}{2} d((\vec{r} \cdot \vec{B}) \vec{r}) - (\vec{r} \cdot d\vec{r}) \vec{B} \end{aligned}$$

$$\vec{\tau} = I \oint_L \vec{r} \times (d\vec{r} \times \vec{B})$$

$$= I \oint_L \frac{1}{2} (\vec{r} \times d\vec{r}) \times \vec{B} + \frac{I}{2} \oint_L d((\vec{r} \cdot \vec{B}) \vec{r})$$

$$- I \vec{B} \oint_L \vec{r} \cdot d\vec{r}$$

$$= I \oint_L \frac{1}{2} (\vec{r} \times d\vec{r}) \times \vec{B} + \frac{I}{2} \oint_L d((\vec{r} \cdot \vec{B}) \vec{r}) - I \vec{B} \oint_L d(\frac{1}{2} r^2)$$

$$= \vec{m} \times \vec{B}$$

对于非均匀磁场，可以证明 $\vec{F} = (\vec{m} \cdot \nabla) \vec{B}$

$$\vec{\tau} = \vec{m} \times \vec{B} + \vec{r} \times (\vec{m} \cdot \nabla) \vec{B}$$

三、对电流作用时的等效外场。

应用 $\vec{F} = \oint I d\vec{l} \times \vec{B}$ 和 $\vec{\tau} = \oint \vec{r} \times (I d\vec{l} \times \vec{B})$ 计算电流体系所受的力和力矩，应先求出外磁场 $\vec{B}$ 。

一般情况下，由外磁场 $\vec{B}$ 和受作用电流的磁场 $\vec{B}_1$ 合成的总磁场的磁感应强度 $\vec{B}_t$ 容易计算。

$$\vec{B}_t = \vec{B} + \vec{B}_1$$

若所考察的受作用对象是稳恒的闭合电流（体电流/面电流），可以证明内力相互抵消，从而可以用 $\vec{B}_t$ 计算受力。

回顾，受力带电体内力相互抵消的证明，

需要用到牛顿第三定律。

困难：线电流元之间的相互作用

$$d\vec{F}_{12} = k \frac{I_2 d\vec{l}_2 \times (I_1 d\vec{l}_1 \times \vec{r}_{12})}{r_{12}^3}$$

不遵守牛顿第三定律

破解：Ampère 定律的原始形式（遵守牛顿第三定律）

$$d\vec{F}_{12} = -k I_1 I_2 \vec{r}_{12} \left[\frac{2}{r_{12}^3} (d\vec{l}_1 \cdot d\vec{l}_2) - \frac{3}{r_{12}^5} (d\vec{l}_1 \cdot \vec{r}_{12})(d\vec{l}_2 \cdot \vec{r}_{12}) \right]$$

$$d\vec{F}_{12} - d\vec{F}_{21} = d \left[k I_1 I_2 \frac{(d\vec{l}_2 \cdot \vec{r}_{12}) \vec{r}_{12}}{r_{12}^3} \right]$$

⇒ 两公式对闭合线圈有相同结果。可等效替换。

因此，

$$\oint I d\vec{l} \times \vec{B}_1 = \oint I d\vec{l} \times \frac{\mu I}{4\pi} \oint \frac{d\vec{l}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

$$= \frac{\mu I^2}{4\pi} \oint \oint \frac{d\vec{l} \times [d\vec{l}' \times (\vec{r} - \vec{r}')] }{|\vec{r} - \vec{r}'|^3}$$

$$= \frac{\mu I^2}{4\pi} \oint \oint d\vec{l}' \frac{d\vec{l} \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} - \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} (d\vec{l} \cdot d\vec{l}')$$

$$= \frac{\mu I^2}{4\pi} \left[\oint d\vec{l}' \oint d\vec{l} \cdot \nabla \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) - \oint \oint \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} (d\vec{l} \cdot d\vec{l}') \right]$$

$$\vec{r} \leftrightarrow \vec{r}' \quad \Rightarrow 0$$

$$\oint \oint \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} (d\vec{l} \cdot d\vec{l}') \xrightarrow{\vec{r} \leftrightarrow \vec{r}'} \oint \oint \frac{\vec{r}' - \vec{r}}{|\vec{r}' - \vec{r}|^3} (d\vec{l}' \cdot d\vec{l}) = - \oint \oint \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} (d\vec{l} \cdot d\vec{l}')$$

$$\Rightarrow \oint \oint \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} (d\vec{l} \cdot d\vec{l}') = 0.$$

$$\Rightarrow \oint I d\vec{l} \times \vec{B}_1 = 0$$

$$\Rightarrow \vec{F} = \oint I d\vec{l} \times \vec{B} = \oint I d\vec{l} \times (\vec{B}_t - \vec{B}_1) = \oint I d\vec{l} \times \vec{B}_t$$

注：上述分析对线电流不成立，因为一个闭合线电流中的任意一小段线电流所受的力趋于 ∞ 。

四、多种载流子的 Hall 效应。

存在两种载流子时的 Hall 系数。

难点：由于 $E = v_1 B$ 、 $E = v_2 B$ 无法同时满足

模仿单一载流子的推导不可行。

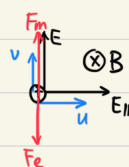
破解：横向受力为 0 → 横向电流为 0。

设 $\vec{E}_0$ 为单位电荷非静电力。

$$\vec{j} = \sigma(\vec{E} + \vec{E}_0) = nq\vec{u}$$

$$\Rightarrow \vec{u} = \frac{\sigma}{nq}(\vec{E} + \vec{E}_0) = \mu(\vec{E} + \vec{E}_0)$$

其中，迁移率 $\mu = \frac{\sigma}{nq} = \frac{u}{E + E_0}$ 为漂移速度与单位电荷驱动力之比



纵向电场驱动 u_1, u_2

$$u_1 = \mu_1 E_{||}$$

$$u_2 = \mu_2 E_{||}$$

单位电荷横向力为 $u_1 B + E = \mu_1 E_{||} B + E$

$$u_2 B + E = \mu_2 E_{||} B + E$$

驱动横向速度 $u_1 = \mu_1 (\mu_1 E_{||} B + E)$

$$u_2 = \mu_2 (\mu_2 E_{||} B + E)$$

横向电流为 0

$$0 = n_1 q_1 u_1 + n_2 q_2 u_2$$

$$= n_1 q_1 \mu_1 (\mu_1 E_{||} B + E) + n_2 q_2 \mu_2 (\mu_2 E_{||} B + E)$$

$$\Rightarrow E = - \frac{(n_1 q_1 \mu_1^2 + n_2 q_2 \mu_2^2) E_{||} B}{n_1 q_1 \mu_1 + n_2 q_2 \mu_2}$$

纵向电流密度

$$j = n_1 q_1 u_1 + n_2 q_2 u_2 = (n_1 q_1 \mu_1 + n_2 q_2 \mu_2) E_{||}$$

$$R_H = \frac{|E|}{B j} = \frac{n_1 q_1 \mu_1^2 + n_2 q_2 \mu_2^2}{(n_1 q_1 \mu_1 + n_2 q_2 \mu_2)^2}$$

$$\text{推广: } R_H = \frac{\sum_i n_i q_i \mu_i^2}{(\sum_i n_i q_i \mu_i)^2}$$

* 更严格的推导

$$\vec{E} = E \hat{x} \quad \vec{B} = B \hat{z}$$

$$\vec{v}_1 = v_{1x} \hat{x} + v_{1y} \hat{y} \quad \vec{v}_2 = v_{2x} \hat{x} + v_{2y} \hat{y}$$

$$\vec{v}_1 = \frac{\mu_1}{q_1} (q_1 E \hat{x} + q_1 \vec{v}_1 \times B \hat{z} + q_1 E_{||} \hat{y})$$

$$\vec{v}_2 = \frac{\mu_2}{q_2} (q_2 E \hat{x} + q_2 \vec{v}_2 \times B \hat{z} + q_2 E_{||} \hat{y})$$

$$\begin{cases} v_{1x} = \mu_1 E + \mu_1 v_{1y} B \\ v_{1y} = -\mu_1 v_{1x} B + \mu_1 E_{||} \\ v_{2x} = \mu_2 E + \mu_2 v_{2y} B \\ v_{2y} = -\mu_2 v_{2x} B + \mu_2 E_{||} \end{cases} \Rightarrow \begin{cases} v_{1x} = \frac{\mu_1(E + \mu_1 B E_{||})}{1 + \mu_1^2 B} \\ v_{1y} = \frac{\mu_1(E_{||} - \mu_1 B E)}{1 + \mu_1^2 B} \\ v_{2x} = \frac{\mu_2(E + \mu_2 B E_{||})}{1 + \mu_2^2 B} \\ v_{2y} = \frac{\mu_2(E_{||} - \mu_2 B E)}{1 + \mu_2^2 B} \end{cases}$$

横向电流

$$0 = j_{||} = n_1 q_1 v_{1y} + n_2 q_2 v_{2y}$$

$$= n_1 q_1 \frac{\mu_1(E_{||} - \mu_1 B E)}{1 + \mu_1^2 B} + n_2 q_2 \frac{\mu_2(E_{||} - \mu_2 B E)}{1 + \mu_2^2 B}$$

$$\Rightarrow \left(\frac{n_1 q_1 \mu_1}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2}{1 + \mu_2^2 B} \right) E_{||} = \left(\frac{n_1 q_1 \mu_1^2}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2^2}{1 + \mu_2^2 B} \right) B E$$

纵向电流

$$j = n_1 q_1 v_{1x} + n_2 q_2 v_{2x}$$

$$= n_1 q_1 \frac{\mu_1(E + \mu_1 B E_{||})}{1 + \mu_1^2 B} + n_2 q_2 \frac{\mu_2(E + \mu_2 B E_{||})}{1 + \mu_2^2 B}$$

$$= \left(\frac{n_1 q_1 \mu_1}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2}{1 + \mu_2^2 B} \right) \left(E + \frac{\frac{n_1 q_1 \mu_1^2}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2^2}{1 + \mu_2^2 B}}{\frac{n_1 q_1 \mu_1}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2}{1 + \mu_2^2 B}} B E_{||} \right)$$

$$= \left(\frac{n_1 q_1 \mu_1}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2}{1 + \mu_2^2 B} \right) E \left(1 + \frac{\left(\frac{n_1 q_1 \mu_1^2}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2^2}{1 + \mu_2^2 B} \right)^2 B^2}{\left(\frac{n_1 q_1 \mu_1}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2}{1 + \mu_2^2 B} \right)^2} \right)$$

$$\approx (n_1 q_1 \mu_1 + n_2 q_2 \mu_2) E$$

$$E_{||} = \frac{\frac{n_1 q_1 \mu_1^2}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2^2}{1 + \mu_2^2 B}}{\frac{n_1 q_1 \mu_1}{1 + \mu_1^2 B} + \frac{n_2 q_2 \mu_2}{1 + \mu_2^2 B}} B E$$

$$\approx \frac{n_1 q_1 \mu_1^2 + n_2 q_2 \mu_2^2}{n_1 q_1 \mu_1 + n_2 q_2 \mu_2} B E$$

$$R_H = \frac{E_{||}}{j B} \approx \frac{n_1 q_1 \mu_1^2 + n_2 q_2 \mu_2^2}{(n_1 q_1 \mu_1 + n_2 q_2 \mu_2)^2}$$

五. 顺磁效应理论.

$$0 \sim \theta + d\theta \text{ 的分子数 } dn(\theta) = n_0 \cdot \frac{2\pi \sin \theta d\theta}{4\pi}$$

分子磁矩取向服从 Boltzmann 分布

$$dn(\theta) = C e^{-\frac{\epsilon_p}{k_B T}} \sin \theta d\theta$$

$$\epsilon_p = -\vec{m}_0 \cdot \vec{B} = -m_0 B \cos \theta$$

$$dn(\theta) = C e^{\frac{m_0 B}{k_B T} \cos \theta} \sin \theta d\theta$$

$$\Rightarrow n_0 = \int_0^\pi C e^{\frac{m_0 B}{k_B T} \cos \theta} \sin \theta d\theta$$

$$\approx \int_0^\pi C \left(1 + \frac{m_0 B}{k_B T} \cos \theta \right) \sin \theta d\theta$$

$$\Rightarrow C = \frac{n_0}{2}$$

$$\Rightarrow dn(\theta) = \frac{n_0}{2} \left(1 + \frac{m_0 B}{k_B T} \cos \theta \right) \sin \theta d\theta$$

$$M = \int_0^\pi m_0 \cos \theta dn(\theta) = \frac{m_0 n_0}{2} \int_0^\pi \cos \theta \left(1 + \frac{m_0 B}{k_B T} \cos \theta \right) \sin \theta d\theta$$

$$= \frac{n_0 m_0^2}{3 k_B T} B \approx \frac{\mu_0 n_0 m_0^2}{3 k_B T} H$$

$$\star \vec{M} = \frac{\mu_0 n_0 m_0^2}{3 k_B T} \vec{H} \quad \chi_m = \frac{\mu_0 n_0 m_0^2}{3 k_B T} \quad \text{Curie 定律.}$$

问题 1: 实验表明, 上式对气态顺磁质适用, 对某些液态和固态顺磁质不成立, 原因何在?

线索 1: 气态适用, 气态与液态和固态的明显区别在于分子间距足够大.

线索 2: 抗磁质的微观理论比实际符合得很好.

抗磁效应比顺磁效应弱.

结论: 当分子固有磁矩足够大且间距很小时, 分子间的磁矩的相互作用不可忽略.

相似的例子: 理想气体、van der Waals 气体.

问题 2: 若 $m_0 B \sim k_B T$, 求 M 与 H 的关系.

$$\text{令 } \alpha = \frac{m_0 B}{k_B T}, \text{ 则 } dn(\theta) = C e^{\alpha \cos \theta} \sin \theta d\theta$$

$$n_0 = \int_0^\pi dn(\theta) = \int_0^\pi C e^{\alpha \cos \theta} \sin \theta d\theta = C \frac{e^\alpha - e^{-\alpha}}{\alpha}$$

$$C = \frac{\alpha n_0}{e^\alpha - e^{-\alpha}}$$

$$dn(\theta) = n_0 \frac{\alpha}{e^\alpha - e^{-\alpha}} e^{\alpha \cos \theta} \sin \theta d\theta$$

$$M = \int_0^\pi m_0 \cos \theta dn(\theta) = \frac{n_0 m_0 \alpha}{e^\alpha - e^{-\alpha}} \int_{-1}^1 x e^{\alpha x} dx$$

$$= \frac{n_0 m_0 \alpha}{e^\alpha - e^{-\alpha}} \left(\frac{x}{\alpha} e^{\alpha x} - \frac{1}{\alpha^2} e^{\alpha x} \right) \Big|_{-1}^1$$

$$= n_0 m_0 \left(\frac{e^\alpha + e^{-\alpha}}{e^\alpha - e^{-\alpha}} - \frac{1}{\alpha} \right)$$

$$= n_0 m_0 \left(\coth \alpha - \frac{1}{\alpha} \right)$$

$$\Rightarrow M = n_0 m \left(\coth \alpha - \frac{1}{\alpha} \right) = \chi_m H$$

$$\alpha \rightarrow 0. \quad M = \frac{1}{3} n_0 m_0 \alpha = \frac{n_0 m_0^2}{3 k_B T} B \approx \frac{\mu_0 n_0 m_0^2}{3 k_B T} H \quad (\text{线性顺磁化})$$

$$\alpha \rightarrow \infty. \quad M = n_0 m_0 \quad (\text{饱和顺磁化})$$

六、分区均匀介质

1. 介质界面与B线重合

传导电流(在真空中)产生的磁感应强度 $\vec{B}_0$

待求静磁场的磁感应强度 $\vec{B} = \frac{\vec{B}_0}{\mu_0}$

磁感应强度 $\vec{B}_1 = \mu_1 \vec{H} = \frac{\mu_1}{\mu_0} \vec{B}_0$

2. 介质界面与B线垂直

介质的引入不改变磁感应强度的分布形式，改变的仅仅是其强度。

真空中磁感应强度 $\vec{B}_0$ 。

待求磁感应强度 $\vec{B} = \alpha \vec{B}_0$

$$\oint_L \frac{\vec{B}}{\mu} \cdot d\vec{l} = \sum I_0 \Rightarrow \alpha = \frac{\sum I_0}{\oint_L \frac{\vec{B}_0}{\mu} \cdot d\vec{l}}$$

$$\vec{B} = \alpha \vec{B}_0, \quad \vec{H}_1 = \frac{\vec{B}}{\mu_1}$$

例1. (例6.9)

一无限大平面($z=0$)将磁导率为 μ_1, μ_2 的介质隔开，在y轴上 $y=\pm a$ 的位置分别放置电流强度为 I 的无限长直线电流，电流方向沿x的负向和正向。求空间磁场分布。

解. 去掉介质后，平面 $z=0$ 为磁场的等磁势面。

因此，介质界面垂直于B线。

$$\vec{B}_0 = \frac{\mu_0 I}{2\pi} \left(\frac{z\hat{y} - (y-a)\hat{z}}{(y-a)^2 + z^2} - \frac{z\hat{y} - (y+a)\hat{z}}{(y+a)^2 + z^2} \right)$$

取以 $y=a$ 为圆心的闭合圆回路L

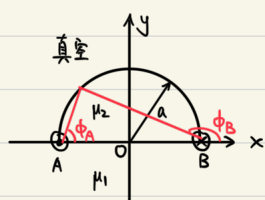
$$\begin{aligned} \oint_L \frac{\vec{B}_0}{\mu} \cdot d\vec{l} &= \frac{1}{\mu_1} \oint_L \vec{B}_0 \cdot d\vec{l} + \frac{1}{\mu_2} \oint_L \vec{B}_0 \cdot d\vec{l} \\ &= \left(\frac{1}{\mu_1} + \frac{1}{\mu_2} \right) \frac{1}{2} \oint_L \vec{B}_0 \cdot d\vec{l} \\ &= \left(\frac{1}{\mu_1} + \frac{1}{\mu_2} \right) \frac{1}{2} \mu_0 I \end{aligned}$$

$$\Rightarrow \alpha = \frac{\sum I_0}{\oint_L \frac{\vec{B}}{\mu} \cdot d\vec{l}} = \frac{2\mu_1 \mu_2}{\mu_0 (\mu_1 + \mu_2)}$$

$$\Rightarrow \vec{B} = \alpha \vec{B}_0 = \frac{\mu_1 \mu_2 I}{\pi (\mu_1 + \mu_2)} \left(\frac{z\hat{y} - (y-a)\hat{z}}{(y-a)^2 + z^2} - \frac{z\hat{y} - (y+a)\hat{z}}{(y+a)^2 + z^2} \right)$$

例2. (习题6.14)

一半无限大磁导率为 μ_1 的磁介质表面放一磁导率为 μ_2 的无限长磁介质圆柱，半径为 a 。设在A、B两处放入反向直线电流 I 。求流方向与圆柱轴平行。求空间磁感应强度分布



首先， $y=0$ 平面与B线垂直。

然后，分析半圆 $r=a$ 。

无限长直线电流

$$\vec{H} = \frac{I}{2\pi r} \hat{\phi} = -\frac{1}{r} \frac{\partial \varphi_m}{\partial \phi} \hat{\phi}$$

$$\Rightarrow \varphi_m = -\frac{I}{2\pi} \phi + C$$

$$\text{半圆上 } \varphi_m = -\frac{I}{2\pi} \phi_A - \frac{I}{2\pi} \phi_B = \frac{I}{2\pi} (\phi_B - \phi_A) = \frac{I}{2\pi} \cdot \pi = \frac{I}{2}$$

因此，半圆与等磁势面重合，与B线垂直。

$$\vec{B}_0 = \frac{\mu_0 I}{2\pi} \left(\frac{-y\hat{x} + (x+a)\hat{y}}{(x+a)^2 + y^2} - \frac{-y\hat{x} + (x-a)\hat{y}}{(x-a)^2 + y^2} \right)$$

取以A为圆心，半径 $\rho \rightarrow 0$ 的无限小回路。

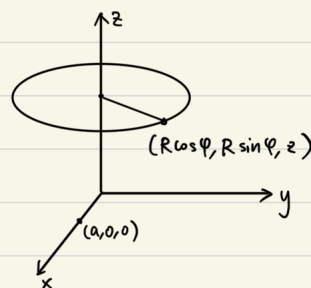
$$\begin{aligned} \oint_L \frac{\vec{B}_0}{\mu} \cdot d\vec{l} &= \frac{1}{\mu_1} \oint_L \vec{B}_0 \cdot d\vec{l} + \frac{1}{\mu_2} \oint_L \vec{B}_0 \cdot d\vec{l} + \frac{1}{\mu_0} \oint_L \vec{B}_0 \cdot d\vec{l} \\ &= \left(\frac{2}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_0} \right) \frac{1}{2} \oint_L \vec{B}_0 \cdot d\vec{l} \end{aligned}$$

$$= \left(\frac{2}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_0} \right) \frac{1}{2} I$$

$$\alpha = \frac{I}{\oint_L \frac{\vec{B}}{\mu} \cdot d\vec{l}} = \frac{4\mu_0 \mu_1 \mu_2}{2\mu_0 \mu_2 + \mu_0 \mu_1 + \mu_1 \mu_2}$$

$$\vec{B} = \frac{2\mu_0 \mu_1 \mu_2}{\pi (2\mu_0 \mu_2 + \mu_0 \mu_1 + \mu_1 \mu_2)} \left(\frac{-y\hat{x} + (x+a)\hat{y}}{(x+a)^2 + y^2} - \frac{-y\hat{x} + (x-a)\hat{y}}{(x-a)^2 + y^2} \right)$$

七、理想螺线管内外磁场。



建立合适的坐标系。

$$\begin{aligned} \vec{B} &= \frac{\mu_0}{4\pi} \int_{-\infty}^{+\infty} dz \int_0^{2\pi} n I d\phi \times \frac{\vec{r}}{r^3} \\ &= \frac{\mu_0}{4\pi} \int_{-\infty}^{+\infty} dz \int_0^{2\pi} n I R d\phi \hat{\phi} \times \frac{(a-R\cos\phi)\hat{x} + (0-R\sin\phi)\hat{y} - z\hat{z}}{[(a-R\cos\phi)^2 + (R\sin\phi)^2 + z^2]^{3/2}} \end{aligned}$$

$$= \frac{\mu_0 n I R}{4\pi} \int_0^{2\pi} d\phi \int_{-\infty}^{+\infty} dz \frac{(R-a\cos\phi)\hat{x} - z\hat{z}}{(a^2 + R^2 - 2aR\cos\phi + z^2)^{3/2}}$$

$$= \frac{\mu_0 n I R}{4\pi} \int_0^{2\pi} d\phi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{R-a\cos\phi}{a^2 + R^2 - 2aR\cos\phi} \cos\alpha d\alpha \hat{x}$$

$$= \frac{\mu_0 n I R}{2\pi} \int_0^{2\pi} \frac{R-a\cos\phi}{a^2 + R^2 - 2aR\cos\phi} d\phi \hat{x}$$

$$= \frac{\mu_0 n I}{2\pi} \hat{x} \int_{-\infty}^{+\infty} \frac{R(R-a\frac{1-u^2}{1+u^2})}{a^2 + R^2 - 2aR\frac{1-u^2}{1+u^2}} \frac{2}{1+u^2} du$$

$$= \frac{\mu_0 n I}{2\pi} \hat{x} \int_{-\infty}^{+\infty} \left(\frac{1}{1+u^2} + \frac{R^2 - a^2}{(R-a)^2 + (R+a)^2 u^2} \right) du$$

$$= \frac{\mu_0 n I}{2\pi} \hat{x} \left(\int_{-\infty}^{+\infty} \frac{1}{1+u^2} du \pm \int_{-\infty}^{+\infty} \frac{ds}{1+s^2} \right)$$

$$= \begin{cases} \mu_0 n I & a < R \\ 0 & a > R \end{cases}$$